

Understanding Cumulative Distribution Functions

Instructions: Work in groups. Show all reasoning and be prepared to explain your answers. Each group should justify conclusions clearly — not just compute numbers.

Group A Problem: Is This a Valid CDF?

A function $F(x)$ is defined by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \leq x < 1 \\ 0.8 & 1 \leq x < 3 \\ 1.1 - \frac{1}{x} & x \geq 3 \end{cases}$$

1. Is $F(x)$ a valid cumulative distribution function?
2. If not, identify **every** property of a CDF that fails.
3. If not, modify the function so that it *becomes* a valid CDF.
4. Describe (in words) what kind of random variable this represents.

Group B Problem: Is This Really a CDF?

Suppose someone claims the following function is a cumulative distribution function:

$$F(x) = \begin{cases} 0 & x < -1 \\ 0.2 & -1 \leq x < 0 \\ 0.2 + \frac{x^2}{4} & 0 \leq x < 2 \\ 0.8 & 2 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

1. Is this a valid CDF? Why? Justify your answer carefully.
2. If not, identify exactly which CDF properties fail.
3. If not, modify the function so that it becomes a valid CDF.
4. For your version or the original version:
 - (a) Find all point masses.
 - (b) Find the density where it exists.
 - (c) Classify the distribution.

Group C Problem: Designing a CDF Under Constraints

Construct a CDF $F(x)$ that satisfies:

- $P(X = 0) = 0.3$
- $P(0 < X < 2) = 0.4$
- $P(X \geq 2) = 0.3$
- X has a continuous density on $(0, 2)$